

PAPER 3 — HANDOFF v12.0

Written 3 September 2026. This is the newest handoff — read it first, and where any earlier document disagrees with it, this one wins.

A note on the numbering, because it drifted on disk. The handoffs are counted by FILE name, and the version printed inside some of the older files does not match. In particular `PAPER3_HANDOFF_v11_2.pdf` is internally titled "v13.0". To avoid confusion, this document refers to the earlier handoffs by their file names only.

Uploaded alongside this one, kept for history and not repeated here in full:

- `PAPER3_HANDOFF_v11_2.pdf` — the immediate predecessor. Shape rules (`contact_band_mm`), the scaled-cuboid primitive, shape-keyed calibration, the workflow for a new object, the six limits of `design_grid`.
- `PAPER3_HANDOFF_v10_0.md` — Block 2 / Block 3 and the diagonal-blob finding.
- `PAPER3_HANDOFF_v9_0.pdf` — the oblique-contact characterisation and the motion-pipeline repairs.

Two earlier files are referenced by a couple of measurements below but are NOT required: `v8_0` (the compliance-limited contact width) and `v12_0` (contact-aware closing, the object library, the batch runner). Every number taken from them is restated here so nothing depends on having them to hand.

0. READ THIS FIRST — THERE IS A RUN IN FLIGHT

Two sweeps were started at the end of this session and are **not yet analysed**. The whole point of this handoff is that the next session can pick them up cleanly.

Run	Object	Points	coarse interior
Cuboid	30 x 120 x 140 scaled cuboid	64	ON, N=2, band -11
Cylinder	Ø26 x 140	45	OFF

Everything else is identical between them: contact calibration file, fixed-angle closing, 6 mm step, `GRASP_TOOL_COLLISION=1`. **Shape is the only variable.**

What to do when Kourosh shares the results

He will run, for each sweep: **Stitch → Grid Accuracy → Grid Quality → Save Run Manifest**, and share:

- `gui_config_used.json`
- `execution_ledger.json`
- `stitch_report.txt`, `stitched_sl.png`
- `grid_accuracy.txt`, `grid_quality_sl.txt`
- `run_manifest.txt`
- two heatmaps: `pt00` and one edge-straddling point
- one training-pair `.npz` from an edge anchor

Check these five things, in this order:

- Grid Quality — zero BLOCKED points, indentation flat across all rows.** Expect 1.2–1.5 mm everywhere. If any row is low, the throat fix did not hold; go to §4.
- Grid Accuracy — scatter under 1% of a taxel.** The last measured run was 0.17%. If it jumps, the arm or the placement chain changed.
- Object outline and EXPECTED panel match the real shape** in both the heatmap 4th panel and the stitched 3rd panel. A 30 mm-wide outline on a 120 mm cuboid means the config carries a stale `shape`; see §5.1.
- Stitched edge lands where the geometry says.** On the cuboid, contact should die out past world Y = 259 (the face edge at object centre 199 + 60). On the cylinder, past the silhouette.
- Training pair alignment** — the `.npz` input frame must sit inside the target map at the right offset.

If all five pass, **the collection pipeline is cleared for both shapes** and the project moves to Block 3 (§8).

1. BIG PICTURE (unchanged)

Kourosh Jolaei, PhD candidate, CoRo Lab, ÉTS Montréal. Supervisors Vincent Duchaine and Jean-Philippe Roberge.

- Paper 1** (J. Robotics and Mechatronics 38(3), 2026) — shape-classifier + hand-crafted extrapolation. 94.3% held-out, 80.8% unseen real objects.
- Paper 2** (submitted to Robotics and Autonomous Systems) — tactile-guided regrasping, CDT paradigm, 602 trials.
- Paper 3** (this work) — **replace the classifier + hand-crafted extrapolation with a learned contact-completion model** trained on large-scale Isaac Sim data.

Four-block plan. Block 1 data collection: complete. Block 2 stitching: complete and validated. Block 3 U-Net training: not started. Block 4 A/B against Paper 1's rules: not started.

Stack. UR5e + Robotiq 2F-85, two TSF-85 capacitive pads (7x4 = 28 taxels each), Isaac Sim 5.1, TSF-85 extension by Berith Atemoztli De la Cruz Sánchez, cuRobo. Ubuntu 22.04.

2. WHERE THE PROJECT IS

Area	State
Cylinders	Widths 18, 26, 40, 60 validated. Ø26 x 140 re-sweep in flight.
Cuboids	30 x 120 x 140 validated end to end. 64-point sweep in flight.
Spheres	Untouched. Deliberately deferred — see §9.4.
Rolled / tilted data	Still blocked on Berith's CNN. Collect 0° and 90° only.
Grid thinning	New this session. Cuboids only. See §3.
Run provenance	New this session. <code>run_manifest.py</code> . See §6.

Verify file freshness before editing

File	Signature to grep for
<code>main_gui.py</code>	<code>coarsen_interior / coverage_holes / do_save_manifest</code>
<code>viz/grid_quality.py</code>	<code>BLOCK_SHORTFALL_RAD</code>
<code>viz/grid_accuracy.py</code>	<code>scatter_taxel_frac</code>
<code>viz/run_manifest.py</code>	<code>resolve_calibration</code>
<code>viz/blob_axis.py</code>	<code>_flat = str(scene.get</code>
<code>sim/collect_from_config.py</code>	<code>SCALED CUBOID / cal_key</code>

3. NEW THIS SESSION: GRID THINNING (`coarse interior`)

3.1 The idea

On a 30 x 120 x 140 cuboid, 45 of 117 grid points sit with the pad wholly on flat face. Their training pairs are trivial — plateau in, plateau out. But they are **not waste**: the extended target map for an EDGE anchor is built from its neighbours, so deleting the interior outright truncates exactly the maps that matter. So thin it, don't remove it.

3.2 The rule is shape-dependent

Shape	Thins	"Edge" test
cuboid / cube	ACROSS + ALONG	pad vs the face silhouette
cylinder	ALONG only	contact strip vs the pad edge
sphere	nothing — returns unchanged and says why	—

Why the cylinder differs. Contact on a rod is a narrow strip down the middle of the pad, compliance-limited to half-width `sqrt(2*R*delta)` — **not** the silhouette. Stepping ACROSS moves that strip to a different place on the pad, so every column reads differently and nothing across may be thinned. Stepping ALONG repeats the identical strip, so ALONG is the redundant axis.

3.3 Settings that are known-good

Cuboid, coarse interior ON: `keep 1 interior point in N = 2`, `edge band = -11`.

- N must be 2. N=3 and N=4 tear holes in the canvas **at every band** — measured, 4.1–7.4% of cells uncovered. The tear does not open inside the interior; it opens where the thinned lattice meets the full-resolution edge band, because the two do not line up.
- band -11 is one pad half-width: "keep it only if the pad is at least HALF off the object".

Measured on the 30 x 120 x 140:

band	N=2	N=3	N=4
+0.0	87	78 (holes)	76 (holes)
-4.0	72	65 (holes)	61 (holes)
-11.0	64	48 (holes)	46 (holes)

64 points, **zero uncovered canvas cells**, 49 straddling an edge. The full grid is 117 points, also zero holes, 72 straddling.

Cylinder: leave it OFF, at every diameter. On Ø26 it saves 4 of 45 points. The useful across range is only ±12 mm and the strip is ±6 mm, so nearly every position already counts as an edge — there is no plateau to thin. Keep the capability for a *long* cylinder later, where the along axis has many interior rows.

3.4 Always confirm coverage before saving

The preview line now reads either **"coverage complete"** or a red **"△ N canvas cells have NO grasp"**. `coverage_holes()` rasterises every footprint onto a 1 mm canvas and counts cells nobody reaches. Spacing arithmetic is not a substitute: 18 mm steps under a 22 mm pad "should" overlap and still tear.

4. NEW THIS SESSION: THE THROAT JAM (fixed)

4.1 The fault

On `run_20260901_154153` (117-point cuboid), the ledger said 117/117 executed, 0 dead, `exec_stage: complete` on every point. It looked perfect. **The entire bottom row of the grid grasped at 40% of the intended squeeze.**

z row	close_rad stopped	indentation
21.69	0.47434	0.53 mm
27.69 – 69.69	0.49100 – 0.49104	0.2341 – 0.4166

Thirteen of 117 points, 16.8 mrad short of the commanded angle where every other row reached it within 0.06 mrad. A physical block. **Nothing caught it** because the ledger's own quality question is `contact_target_reached`, which in fixed-angle mode is `null` by construction — a question that cannot fail. The evidence was in `close_rad_stopped`, unread.

4.2 Three root causes, all "the object is a round rod"

- 1. `load_gripper_cloud(0)` **read the wrong entry.** `finger_throat.json` holds both `"30.0"` (cylinder, pad separation 34.77 mm) and `"30.0|cuboid"` (35.48 mm), and the bare key won. → now shape-keyed, bare key kept as fallback.
- 2. `grid_gripper_cloud_gap_mm` **solved the object as a finite cylinder.** This is the ACTIVE path whenever `finger_throat.json` exists. A 30 x 120 mm box end was tested as a Ø30 rod, shoulders invisible. → now an exact box signed-distance branch. (`_rod_cloud` was fixed too, but that only feeds the fallback path.)
- 3. **The working-radius filter threw away the geometry that mattered.** It dropped every gripper point with `hypot(across, closing) >= pad_separation/2 = 17.74 mm`, on the grounds that anything that far out is a holding surface. **That is true only along the CLOSING axis** — the pads meet there and nowhere else. A gripper part 30 mm out ACROSS the pad is throat. The raw cloud spans ±37.5 mm across; the filter kept 598 of 4944 points. → now tests the closing axis only.

4.3 Evidence for the filter change

Worst gap over the 13 across-columns, margin 1.0 mm:

pad_dz	radial filter	closing-only
+21.69	+4.98	+0.50 → REJECT
+27.69	+10.98	+1.65 pass
+33.69	+16.98	+7.02 pass

It rejects exactly the row that jammed, and only that row. And it does **not** disturb the observations the 1.0 mm margin was bracketed against — all three unchanged to 0.01 mm:

observation	radial	closing-only
Ø60 +29.64 (peak 4, DEAD)	+0.12	+0.12
Ø60 +39.15 (peak 14314, firm)	+3.57	+3.57
Ø26 +16.31 (lowest of six runs)	+1.98	+1.98

The only cylinder pose that moves is Ø26 at +28.31, +13.98 → +5.95, still passing.

4.4 The outcome, which surprised us

After the fix, Design grid returned **117 points again** — but the base rose from `pad_dz 45.69` to **49.29**, moving the bottom row from 21.69 to **25.29**, where the gap is +1.10 mm. `design_grid` re-centres the height first and only drops points if raising cannot fix it. **Judge the fix by the Z values, not the point count.**

4.5 Still open

`probe_finger_throat.py` measures with `rod_radius_mm: 15.0` for the cuboid entry too — the probe still models the object as a rod. The design-time check now uses the real box; the probed depth behind it does not. Not blocking, but the same class of assumption.

5. NEW THIS SESSION: THE EXPECTED PANEL

5.1 Cuboids drew as cylinders (fixed earlier, root cause confirmed)

A run whose `gui_config_used.json` carries `"shape": "cylinder"` and no `across_mm` makes every consumer draw a Ø30 rod around a 120 mm object and predict a 16.3 mm chord band the pad never touches — an empty EXPECTED panel. The GUI writes both fields correctly now. `fix_run_config.py` repairs older run folders; the tactile CSVs were never wrong, so no re-collection is needed.

5.2 Flat faces get a flat profile

`blob_axis.expected_patch_map` applied a Hertzian ridge peaked at the object centre. On a 120 mm face that invented a brightness gradient across the pad and inflated the expected elongation (2.59 against a measured 1.76 on a pose wholly on flat face). Now: `_flat` shapes get uniform pressure wherever pad and face overlap, zero elsewhere. Cylinders keep the Hertzian branch.

5.3 The cylinder band is ~37% too wide — NOT YET FIXED

`heatmaps.py` uses `INDENT_MM = 2.4`, so on Ø26 it prints "band 15.1 mm" = 2.7 taxel columns. **Measured from the real Ø26 pt00 map**, column brightness normalised:

column	0	1	2	3
brightness	0.49	1.00	1.00	0.31

Two columns read full — contact ≈ 11 mm wide, half-width 5.5 mm, implying `delta = half²/2R = 1.16 mm`, the same figure the earliest handoff (v8_0 §3.3) recorded. This is part of why the panel reports expected along 3.31 against a measured 2.79.

Consequence. At `delta = 1.4 mm` a cylinder needs Ø86 for the strip to fill the pad, and the jaw span is 85 mm — **a cylinder can never fully cover the pad in this gripper.** `coarsen_interior` now defaults to `indent_mm = 1.4`. `heatmaps.py` has NOT been changed; doing so would alter the expected panel on every cylinder run, so it is a deliberate decision left to the next session.

6. NEW THIS SESSION: REPORTING AND PROVENANCE

Three new modules in `viz/`, each with a button on the Stitching tab.

6.1 `grid_quality.py` — what happened at every grid point

Per point: indentation at hold, close-angle shortfall (`ceiling - stopped`), tactile peak sum, verdict (OK, POOR, 1/2-BLOCKED, DEAD, 20% TARGET MISSED). Aggregated **by Z row and by Y column**, which is what makes a positional fault obvious — a fault invisible in a 117-line list is unmissable once

grouped.

Thresholds are relative to the run's own medians, because a Ø18 cylinder and a 40 mm cuboid stall at completely different angles: `BLOCK_SHORTFALL_RAD = 0.005` (good points scatter by 0.06 mrad, the bad row was 16.8 away), `WEAK_DEFORM_FRAC = 0.70`.

In fixed-angle mode the shortfall is the only evidence that a close was obstructed. Read it first.

6.2 `grid_accuracy.py` — **commanded vs measured pad centre**

Rewritten this session into four panels, each titled with the question it answers: WHERE is the error (spatial map, colour = size, arrows = direction at a printed magnification); HOW BIG is it really (drawn inside one taxel at true scale, with a magnified inset); Does it DRIFT (versus visit order); Any bad ROW or COLUMN.

Reports everything as a **fraction of one taxel** (5.5 x 5.29 mm), because millimetres need a division before they mean anything.

Last measured (117-point cuboid): bias dY -0.005, dZ +0.029 mm; scatter dY 0.009, dZ 0.007 mm = **0.17% of a taxel**; worst miss 0.074 mm = 1.33%. Bias translates the whole map and does not blur it; scatter is what smears a stitch.

6.3 `run_manifest.py` — **make a run folder self-contained**

Inventories the folder against 17 expected items and flags anything required-and-missing. Copies the calibration and throat entries **actually used** into `Provenance/`. Dumps every derived number as JSON alongside the prose. Hashes nine source files so two runs produced by different code cannot be pooled unnoticed.

It identifies the calibration file by measurement, not by trusting a flag. `GRASP_CAL_READ` is never written into the run folder. But `collision_model.pad_offset_mm` is `T00L_OFFSET_Z` and `close_rad_ceiling` is `close_rad`, so the entry matching both is the one that was read. On the 117-point run: key `30.0|cuboid` exists in *both* the fixed and contact files with the same `T00L_OFFSET_Z` 0.15581, discriminated purely by `close_rad` — 0.4931 contact vs 0.519 fixed. Resolved to **contact**, unambiguously.

7. CLOSING MODE — SETTLED, DO NOT RE-OPEN

A spread of 1.10–1.60 mm indentation across nine geometrically similar points looked like a closing-mode problem. **It was measured and it is not.** Contact-aware closing was run on the identical grid. All nine points came back `contact_target_reached: false` and `close_rad_stopped` was the *same* 0.49102 as fixed-angle. The collector's own log says it: *"TARGET NEVER REACHED ... Closed to the ceiling — i.e. exactly what the fixed-angle mode would have done."*

	fixed angle	contact-aware
indentation at hold	1.10 – 1.60 mm	1.13 – 1.50 mm
overlap sigma s1	112.4 (41%)	109.4 (40%)
overlap sigma s2	120.9 (47%)	120.0 (47%)

Why it cannot trip. The fixed angle is a hard ceiling by design: contact-aware can only stop *earlier*, never squeeze further. Collection's ceiling is set to `close_rad` = the angle at which calibration *reached* the target (0.4931), leaving no headroom. And `contact_at_close` is ~1.1 mm while `deformation_raw_at_hold` is ~1.4 mm — the pad settles through the 3.5 s hold, so a target validated against the hold value is unreachable at close time.

Decisions. Keep fixed-angle. Do not raise the ceiling. Do not wire a Collection-tab contact-close control.

Three things this bought: squeeze variation is ruled out as the source of overlap sigma (the row-gain artifact is now the only candidate left standing); an unintentional repeatability test passed well (blob +2.2 vs +2.0°, elong 2.91 vs 2.96, peak sums 10702 vs 10002); and if per-point squeeze control is ever wanted, the one-line change is to pass the calibration's `contact_close.close_rad_ceiling` (0.519), not its `close_rad`.

8. NEXT STEPS, IN ORDER

8.1 Immediately — close out the two runs in flight

1. Analyse both sweeps against the five checks in §0.
2. If both pass, record in the handoff that **the collection pipeline is cleared for cuboid and cylinder**, and stop polishing it.

8.2 Then — Block 3, and nothing else

This is the honest priority. Over the last several sessions the collection pipeline has been repeatedly improved while Blocks 3 and 4 stayed unstated. Every fix was real and several were serious, but the risk now is not rigour — it is that the dataset tooling keeps absorbing the time.

3. Train the U-Net on the cuboid + cylinder data. The training pipeline exists (`dataset.py`, `model.py`, `train.py`) with masked loss, split-by-run, and baselines against zero and copy-input.
4. Use **edge-straddling anchors**, not pt00. Measured earlier: interior anchors halved TC error, 5.96 → 3.00 mm, crossing Paper 1's 4.42 mm safe-zone threshold.
5. Then Block 4: A/B against Paper 1's shape rules.

8.3 Deferred deliberately

- **Spheres.** Untested end to end; Berith's "spheres" are 40 x 40 x 90 capsules, not spheres. A third shape buys variety that cannot be used while the second is still empty.
- **Rolled / tilted data.** Blocked on the CNN (§9.1).
- `heatmaps.py` **INDENT_MM** (§5.3).
- `probe_finger_throat.py` **rod model** (§4.5).

9. CARRIED FORWARD

9.1 The diagonal blob artifact — still the biggest scientific limitation

Berith confirmed it is real and is working on it. 0° gives a clean vertical ridge and 90° a clean horizontal one, while 20° and 45° collapse into broad blobs — at 45° the map has the HIGHEST sum (12195) with the LOWEST peak (999). **Collect 0° and 90° only.** A completion model trained on maps

where diagonal ridges render as round blobs would learn the renderer's failure, and no downstream metric would reveal it.

Berith is delivering a **new TSF-85 Isaac Sim extension** and later a **Newton port**. This is a new sensor model, not just a scene, so it is the thing that could actually fix the artifact. **The acceptance test is unchanged:** re-run 0/20/45/90 on the new extension and confirm 20° and 45° give elongated ridges before letting any tilted grasp into a training set.

9.2 Known-broken / not-yet-done

- `viz/validation.py` still assumes an axis-aligned pad footprint. Correct on flat runs only.
- **GSR is saturated** — no dynamic range. Do not quote it.
- **No ACROSS-side limit on a hand-typed y**. "auto y anchor" derives a safe value, but a typed y is unchecked; on a 120 mm face, y beyond ~55 runs the pad off entirely.
- **A refused design does not block Save Config**. Design Grid can report REFUSED and the run command is still offered.
- **Cuboid yaw is undefined**. The scene applies no yaw, so the pads meet the +X/-X faces. Correct for a FACE grasp, but nothing records it, and an EDGE or corner grasp is a different tactile signature that Paper 1's rule does not cover.
- **100 mm vs 140 mm bodies** — the earliest sweeps are 140 mm; Berith's objects are 100 mm. Do not pool without saying so.
- `plot_scale.json` **is shared: true, fixed_vmax: null** — each run scales to its own maximum, so brightness is not comparable across runs. Set a fixed max (2400, matching Paper 2's figures) before pooling sweeps or putting two runs side by side in a figure.

9.3 A related paper worth citing

ShapeGrasp: Simultaneous Visuo-Haptic Shape Completion and Grasping, Rustler & Hoffmann, arXiv:2605.02347v2, June 2026 (unrefereed manuscript — cite as such). Iterative grasp-and-complete: complete a 3D shape from one RGB-D view, plan a grasp, feed back tactile contacts **and gripper-occupied "free space"** to refine and regrasp. Same Robotiq 2F-85, tactile points placed via forward kinematics, GSR as the end metric.

Three things to take from it:

1. Their free-space constraint is the argument for **training on no-contact cells rather than masking them** — a zero reading is evidence of absence, not missing data.
2. Their metric set — CD, HD, JS, F1, and especially the **precision/recall split** that separates *hallucinated* from *missing* geometry — is sharper than SSIM alone for Block 4.
3. Their reported failure mode is that tactile input **inflates** the completion. Pre-register that as a check; the learned model inherits the same risk.

9.4 Constants and formulas (unchanged from `v11_2` unless noted)

Name	Value	Provenance
PALM_DROP_MM	86.69	measured: 10.79 + 75.9
THROAT_MARGIN_MM	1.0	bracketed: dead -0.06, D26 lowest +1.88, calibration pose +3.13
DEAD_GRASP_FLOOR	0.0002 m	rest 0.000, firm 1.3–1.5
CONTACT_TARGET	0.0014 m	1.4 mm
across_margin_mm	3.0 mm	pad still touching at the outermost column
INDENT_MM (heatmaps)	2.4 mm	over-reads by ~40%; measured 1.16 on Ø26 — see §5.3
indent_mm (coarsen)	1.4 mm	new, from the Ø26 measurement
canvas_mm	96.0	4 halvings
PAD_W, PAD_H	22.0, 37.0 mm	4 cols x 5.5, 7 rows x 5.29
OBJ_BASE_Z	982.2 mm	obj_z = 982.2 + L/2
grid step default	6.0 mm	0.5 mm of sub-taxel shift per step
contact half-width	<code>sqrt(2*R*delta)</code>	compliance-limited, NOT silhouette

10. FILE INVENTORY

Changed or created this session

File	Change
<code>main_gui.py</code>	shape-aware throat (box SDF on the active path, box cloud on the fallback, shape-keyed lookup, closing-axis filter); <code>coverage_holes</code> ; <code>coarsen_interior</code> + its three GUI controls; <code>_apply_coarse</code> shared by preview and config; grid-quality button + handler + loader; run-manifest button + handler + loader; coarse-band default -11
<code>viz/grid_quality.py</code>	NEW
<code>viz/grid_accuracy.py</code>	rewritten: four question-shaped panels, taxel-relative reporting
<code>viz/run_manifest.py</code>	NEW
<code>viz/blob_axis.py</code>	flat-face expected patch for cuboid/cube
<code>fix_run_config.py</code>	NEW — repairs the object block in an old run's config

Unchanged but relevant

`sim/collect_from_config.py`, `viz/stitching.py`, `viz/heatmaps.py`, `viz/validation.py`, `examples/probe_finger_throat.py`, `Data/build_object_library.py`, `examples/run_batch.py`.

The sweep sequence

Design grid → Save Config → run → Stitch → Grid Accuracy → Grid Quality → Save Run Manifest. The manifest goes last so it captures the other outputs rather than recording them as absent.

11. HOW TO WORK ON THIS

- **One verifiable step at a time. Measure, don't guess. Distances in mm.**
- **Complete files, never diffs. Commands on one line** — multi-line `\` continuations silently drop env vars, and a prefix before `cd` does not survive `&&`.
- **One-line answers per question** unless depth is asked for.
- **Kourosh's visual read of the simulation overrides diagnostics when they conflict.** It has been correct every time.
- **Bad data must refuse LOUDLY rather than degrade silently.**
- Never present schematic pseudocode as a quotable line from an actual file.

The pattern behind almost every fault in this project

A wrong number that looked normal. A hardcoded `"cylinder"` in a config. A grasped width taken as the minimum dimension. A calibration key two shapes shared. A gripper cloud filtered on the wrong axis. A ledger field that could not fail. None of these threw an error.

And one added this session: **a warning that was computed and then discarded.** The first version of `coarsen_interior` detected the coverage holes and stored the message in a variable nothing ever displayed — so the check looked like it had run and passed. That is worse than no check at all. Every guard must print WHAT IT MEASURED, and the answer must reach the screen.

End of handoff v12.0.